Jamshid al-Kashi, Miftah al-Hisab

Islamic Original

Modern LaTeX Editions of Public-Domain Mathematics Manuscripts

بسم الله الرحمن الرحيم

Miftāh al-Ḥisāb

The Key to Arithmetic

مفتاح الحساب

Jamshīd Ghiyāth al-Dīn

al-Kāshī (al-Kāshānī)

d. 832 AH / 1429 CE

A comprehensive encyclopedic treatise on arithmetic, geometry, and algebra

Completed in Samarkand, 2 March 1427 CE (829 AH)

Typeset edition based on the 1887 Arabic edition and modern scholarly translations

Edited with commentary and notes

تحقيق وشرح

Contents

Preface to This Edition

This edition presents the complete structure and representative content of *Miftāh al-Ḥisāb* (The Key of Arithmetic), the crowning achievement of Islamic arithmetic, composed by the renowned Persian mathematician and astronomer Jamshīd Ghiyāth al-Dīn al-Kāshī in 1427 CE.

Al-Kāshī (c. 1380–1429) was the astronomer royal to Ulugh Beg at Samarkand. His *Miftāh al-Ḥisāb* is an encyclopedic work covering three principal subjects: arithmetic, geometry, and algebra. It comprises five treatises on:

1. Arithmetic of integers
2. Arithmetic of fractions (including decimal fractions)
3. Arithmetic of sexagesimal numbers
4. Geometry and measurement
5. Algebra

The work includes al-Kāshī's famous calculation of π to 16 decimal places, his systematic treatment of decimal fractions (the first in history), and methods for root extraction that predate similar European methods by centuries. As Berggren noted, *Miftāh* was "the crowning achievement of Islamic arithmetic, and truly a gift fit for a king."

Note on the text: Arabic passages are set in Noto Naskh Arabic. Mathematical formulas use modern notation while preserving the original algorithms. Tables are adapted for readability.

Author's Preface

بسم الله الرحمن الرحيم
الحمد لله الذي توحد بإبداع الآحاد، وتفرد بتأليف صنوف الأعداد، وصلاة على خير خلقه محمد أشفع الشافعين يوم التناد، وآله وأولاده الهادين إلى سبيل النجاة والرشاد.
أمّا بعد فإن أحوج خلق الله تعالى إلى عفرانه جمشيد بن مسعود بن محمود الطبيب الكاشي الملقب بغياث أحسن أحواله، يقول لما مارست الأعمال الحسابية والقوانين الهندسية حتى بلغت إلى حقائقها، وبالغت في دقائقها، وكشفت غوامضها ومعضلاتها وحللت مشكلاتها، واستنبطت كثيراً من القوانين والضوابط فيها، واستخرجت ما صعب استخراجه على كثير من مباشريها...

Translation:

In the name of God, the Merciful, the Compassionate.

Praise be to God, who is unique in creating units, and peerless in composing the classes of numbers. And blessings upon the best of His creation, Muhammad, the intercessor for the interceders on the Day of Calling, and upon his family and his sons who guide to the path of salvation and right guidance.

Jamshīd ibn Masūd ibn Mahmūd al-Ṭabīb al-Kāshī, called Ghiyāth al-Dīn — may God improve his condition — says:

When I had practiced arithmetic and geometric methods until I reached their true principles, and penetrated their subtleties, and uncovered their hidden difficulties and problems and solved their challenges, and derived many rules and controls therein, and extracted what was difficult to extract for many of their practitioners — as when I recomputed all the tables of the Ilkhanid Zij with the utmost precision, and composed the Zij called the *Khāqānī* completing the Ilkhanid Zij, and collected in it all that I had derived of astronomers' methods that appear in no other zij together with geometric proofs, and composed also the *Zij al-Tashīlāt*, and various tables, and authored other treatises such as the treatise called *Sullam al-Samā'* on resolving difficulties that predecessors had encountered in distances and celestial bodies, and the *Risāla al-Muḥiṭiyya* on the ratio of the diameter to the circumference, and the *Risāla al-Watar wal-Jayb* on extracting the chord and sine for one-third of a given arc, which also was difficult for predecessors — as the author of the *Majisti* said of it that there is no way to achieve it — and I invented the instrument called Ṭabaq al-Manāṭiq, and wrote on its construction and use the treatise *Nuzhat al-Ḥadā'iq*, an instrument by which one may obtain the calendars of the planets and their declinations and distances from Earth and their retrograde motion and eclipses and related phenomena, and solved answers to many questions that skilled calculators asked me by way of examination or learning, and that were not obtained through algebraic methods — I found in the course of these works many rules by which the operations of arithmetic preliminaries may be performed in the easiest manner and the simplest ways and with the least effort and the greatest benefit and the clearest arrangement. So I resolved to record them and to explain them, to serve as a reminder for beloved friends and as guidance for people of understanding.

The author continues, describing his intent to compose a work that would serve as a comprehensive textbook for practitioners of arithmetic, avoiding both tedious excess and abbreviated insufficiency.

Original Table of Contents

فصل في مباني علم الحساب وبيان أسمائه.
المقالة الأولى في حساب الأعداد الصحيحة. وهي سبعة فصول:
المقالة الثانية في حساب الكسور المتصلة والمنفصلة والكسورية.
المقالة الثالثة في حساب الدرج والدقائق والثواني ونحوها.
المقالة الرابعة في حساب المساحات والأحجام.
المقالة الخامسة في الجبر والمقابلة.

The work comprises five treatises:

No.	Treatise
I	**On the Arithmetic of Integers** (حساب الأعداد الصحيحة) — 7 chapters
II	**On the Arithmetic of Fractions** (حساب الكسور) — 10 chapters
III	**On the Arithmetic of Sexagesimal Numbers** (حساب المثالث والدرج) — 10 chapters
IV	**On Measurements and Geometry** (حساب المساحات والأحجام) — 9 chapters
V	**On Algebra** (الجبر والمقابلة) — multiple chapters

In addition, the work contains numerous tables for facilitating computation. As al-Kāshī himself notes:

> I placed for most operations a model in tabular form so that they may be easily grasped by the intelligent.

1 Treatise I: On the Arithmetic of Integers

المقالة الأولى في حساب الأعداد الصحيحة.

This treatise, the foundation of the entire work, comprises **seven chapters** covering all operations on whole numbers:

Ch.1: On the naming of numbers and place values — The Hindu–Arabic numeral system and the principle of place value.

Ch.2: On addition — Methods for summing integers, including column addition.

Ch.3: On subtraction — Techniques for finding differences.

Ch.4: On multiplication — Including the *network method* (الشبكة / *shabaka*), the *method of columns*, and the *method of complement*.

Ch.5: On division — Including long division and abbreviated methods.

Ch.6: On extraction of roots — Square roots and higher-order roots by a general method.

Ch.7: On finding unknowns by algebra — Preliminary applications.

1.1 Chapter 1: On the Naming of Numbers

فصل في التعريف بالأعداد وأسمائها ومنازلها.

Al-Kāshī begins by explaining the decimal place-value system, attributing it to the Indians and noting its transmission to the Islamic world. He describes the names of the place values:

Power	Arabic Name	English
10^0	واحد	Units
10^1	عشرة	Tens
10^2	مئة	Hundreds
10^3	ألف	Thousands
10^4	عشرة آلاف	Ten thousands
10^5	مئة ألف	Hundred thousands
10^6	ألف ألف / مليون	Millions
10^7	عشرة ملايين	Ten millions
10^8	مئة مليون	Hundred millions
10^9	ألف مليون / مليار	Billions (thousand millions)

Al-Kāshī extends this system far beyond what was common in his time, enabling the large-scale calculations that he would later use in his astronomical computations.

1.2 Chapter 4: On Multiplication

فصل في الضرب وطرقه.

Al-Kāshī presents several methods of multiplication. The most famous is the **method of the network** (طريقة الشبكة / *shabaka* or lattice multiplication), which he describes in full detail. He also presents the **method of columns** (طريقة الأعمدة) and methods for mental calculation.

Lattice Multiplication Method:

1. Draw a grid of squares with as many rows and columns as there are digits in the multiplicand and multiplier.
2. Write the multiplicand above the grid and the multiplier to the right.
3. Divide each square diagonally from top-left to bottom-right.
4. Multiply each digit of the multiplicand by each digit of the multiplier, writing the tens above the diagonal and the units below.
5. Sum along each diagonal from bottom-right to top-left, carrying as needed.
6. Read the final product along the left and bottom edges.

Example: To multiply 524×385:

	5	**2**	**4**
3	1/5	0/6	1/2
8	4/0	1/6	3/2
5	2/5	1/0	2/0

Result: $\boxed{524 \times 385 = 201{,}740}$

1.3 Chapter 6: On the Extraction of Roots

فصل في استخراج الجذور.

Al-Kāshī's method for extracting roots is one of the most celebrated parts of the *Miftāh*. He gives a general algorithm for finding the nth root of any number, a method equivalent to what is now known as the **Ruffini–Horner method** (though al-Kāshī predated both by several centuries).

General Method for Extracting the nth Root:
Given a number N and an integer n, to find $\sqrt[n]{N}$:

1. Separate N into groups of n digits from the decimal point.
2. Find the largest integer a such that a^n does not exceed the first group.
3. Subtract a^n from the first group and bring down the next group.
4. Using the binomial expansion of $(a+b)^n$, find the next digit b by dividing the remainder by $n \cdot a^{n-1}$.
5. Iterate until the desired precision is reached.

For the **square root** $(n = 2)$, this reduces to:

Square Root Extraction:

1. Separate the number into pairs of digits from the decimal point.
2. Find the largest square not exceeding the leftmost pair; this gives the first digit of the root.
3. Subtract its square and bring down the next pair.
4. Double the current root, and find a digit such that $(20 \times \text{current} + d) \times d$ does not exceed the remainder.
5. Repeat until all pairs are processed.

Example: $\sqrt{144} = 12$

$$\begin{aligned} 1^2 &= 1 \quad \text{(first digit)} \\ (20 \times 1 + 2) \times 2 &= 44 \\ 44 &= 44 \quad \checkmark \quad \text{(second digit)} \end{aligned}$$

This method for root extraction was transmitted to Europe and appeared in the works of later mathematicians. Al-Kāshī's presentation is remarkably systematic and general, applying to roots of any order.

2 Treatise II: On the Arithmetic of Fractions

المقالة الثانية في حساب الكسور.

This is arguably the most historically significant treatise of the *Miftāh*, for it contains the **first systematic treatment of decimal fractions** in the history of mathematics. As Youschkevitch and Rosenfeld observed: "It is in the work of Giyāth al-Dīn Jamshīd al-Kāshī in the early fifteenth century that we first see both a total command of the idea of decimal fractions and a convenient notation for them."

2.1 The Ten Chapters of Treatise II

Ch.1: On the naming of fractions — Definitions and terminology for common fractions.

Ch.2: On reduction of fractions — Converting to common denominators.

Ch.3: On addition and subtraction of fractions.

Ch.4: On multiplication of fractions.

Ch.5: On division of fractions.

Ch.6: On conversion between fractions — Including sexagesimal to decimal and vice versa.

Ch.7: On the extraction of roots of fractions.

Ch.8: On the *kasr al-a'dad* (fractions of numbers) — A special class of fractions.

Ch.9: On decimal fractions (الكسر العشري) — The crowning achievement.

Ch.10: On the use of decimal fractions in calculations.

2.2 Chapter 9: On Decimal Fractions

الباب التاسع في الكسر العشري واستعماله.

Al-Kāshī introduced decimal fractions with a clarity and generality that was unprecedented. His notation used a vertical line (▯) to separate the integer part from the fractional part. For example, he would write:

$$\boxed{3 \,|\, 14159265\ldots}$$

to represent $3.14159265\ldots$, where the digits after the vertical bar are understood as tenths, hundredths, thousandths, etc.

Al-Kāshī writes:

> Know that the common people call the first place after the units 'the place of the parts of ten' (منزلة أجزاء العشرة), and the second 'the place of the parts of a hundred' (منزلة أجزاء المئة), and the third 'the place of the parts of a thousand' (منزلة أجزاء الألف), and so on, always multiplying by ten. And we have found it appropriate to call the first place after the decimal point 'the place of the first tenth' (منزلة العشر الأول), and the second 'the place of the second tenth' (منزلة العشر الثاني), and so on.

2.2.1 Al-Kāshī's Notation System

Al-Kāshī's notation for decimal fractions was remarkably modern. He wrote the integer part, then a separator, then the fractional digits. For instance:

Al-Kāshī's notation	Modern notation	Value
٧٥٣٩▯١٣	13.7539	Thirteen and seven thousand five hundred thirty-nine ten-thousandths
١٤١٤٢١٣٥▯٠	0.14142135	Approximation of $\frac{1}{\sqrt{2}}$
١٤١٥٩٢٦٥٣٥▯٣	3.1415926535	Approximation of π

2.2.2 Decimal Operations

Al-Kāshī demonstrates all four arithmetic operations with decimal fractions. For multiplication, he gives the rule:

Multiplication of Decimal Fractions:

1. Multiply the two numbers as if they were integers, ignoring the decimal point.
2. In the product, count from the right a number of places equal to the sum of the decimal places in the multiplicand and multiplier.
3. Place the separator at that position.

Example: $0.25 \times 0.4 = 0.10$

- Multiply as integers: $25 \times 4 = 100$
- Total decimal places: $2 + 1 = 3$
- Result: $0.100 = 0.1$

2.2.3 The Significance of Decimal Fractions

Al-Kāshī used decimal fractions extensively in his astronomical calculations. He converted between sexagesimal and decimal systems with ease. As Kennedy observed: "Al-Kāshī was first and foremost a master computer of extraordinary ability, witness his facile use of pure sexagesimals, his wide application of iterative algorithms, and his sure touch in so laying out a computation that he controlled the maximum error and maintained a check at all stages."

Sexagesimal to Decimal Conversion:

To convert a sexagesimal number $N = a + \frac{b}{60} + \frac{c}{60^2} + \cdots$ to decimal:

1. Keep the integer part a as is.
2. Convert each sexagesimal fraction: divide by 60, 60^2, etc.
3. Sum the results in decimal form.

Example: Convert $3; 8, 29, 44$ (sexagesimal) to decimal:

$$3 + \frac{8}{60} + \frac{29}{60^2} + \frac{44}{60^3} = 3.141592\ldots$$

3 Treatise III: On the Arithmetic of Sexagesimal Numbers

المقالة الثالثة في حساب الدرج والدقائق والثواني.

This treatise is devoted to the sexagesimal (base-60) system, which was the standard system for astronomical calculations in the Islamic world. Al-Kāshī writes:

> The astronomers have adopted the sexagesimal system because of the many kinds of divisions it admits, since sixty is divisible by two, three, four, five, six, ten, twelve, fifteen, twenty, and thirty. And because they needed to use fractions in their calculations, they divided the circle into three hundred and sixty degrees, each degree into sixty minutes, each minute into sixty seconds, and so on without end.

3.1 The Chapters of Treatise III

Ch.1: On the naming of sexagesimal places — Degrees (درجة), minutes (دقيقة), seconds (ثانية), thirds (ثالثة), etc.

Ch.2: On addition and subtraction of sexagesimal numbers.

Ch.3: On multiplication of sexagesimal numbers.

Ch.4: On division of sexagesimal numbers.

Ch.5: On the extraction of roots of sexagesimal numbers.

Ch.6: On converting sexagesimal to other systems.

Ch.7: On astronomical calculations.

Ch.8: On operations with the *sine* and *chord*.

3.2 Sexagesimal Notation

Al-Kāshī used the following notation for sexagesimal numbers. In modern notation, we write $a; b, c, d$ to represent:

$$a + \frac{b}{60} + \frac{c}{60^2} + \frac{d}{60^3}$$

Arabic Term	Latin Equivalent	Symbol	Value
درجة	Degree	$^\circ$ or nothing	$60^0 = 1$
دقيقة	Minute	$'$	60^{-1}
ثانية	Second	$''$	60^{-2}
ثالثة	Third	$'''$	60^{-3}
رابعة	Fourth		60^{-4}
خامسة	Fifth		60^{-5}

3.3 Chapter 5: On the Extraction of Roots of Sexagesimal Numbers

This chapter contains al-Kāshī's remarkably general method for extracting nth roots in the sexagesimal system. Dakhel's study of this chapter (1960) showed that al-Kāshī's method is essentially the same as the Ruffini–Horner method.

Sexagesimal Root Extraction:
To find $\sqrt[n]{N}$ in sexagesimal:

1. Express N in sexagesimal form: $N = N_0; N_1, N_2, N_3, \ldots$
2. Group the sexagesimal places appropriately for the root order.
3. Apply the binomial method iteratively:

$$\sqrt[n]{a^n + r} \approx a + \frac{r}{n \cdot a^{n-1}}$$

4. At each step, refine the approximation using the error term.
5. Continue until the desired precision is achieved.

Example — Square Root in Sexagesimal:
Find $\sqrt{2}$ in sexagesimal. Al-Kāshī gives:

$$\sqrt{2} \approx 1; 24, 51, 10, 7, 46, 6, 4, 40, \ldots$$

which corresponds to:

$$1 + \frac{24}{60} + \frac{51}{60^2} + \frac{10}{60^3} + \cdots \approx 1.41421356\ldots$$

This approximation of $\sqrt{2}$ is accurate to the limits of the calculation. Al-Kāshī's method of root extraction is completely general and works for any order of root in any number base.

4 Treatise IV: On Measurements and Geometry

المقالة الرابعة في حساب المساحات والأحجام.

This is the longest and most diverse treatise of the *Miftāh*, covering plane and solid geometry, with applications to surveying, architecture, and engineering. It comprises **nine chapters** with numerous sections.

4.1 Original Table of Contents of Treatise IV

فصل في المساحة وألفاظها.
الباب الأول: في مساحة المثلث وما يتعلق به.
الباب الثاني: في مساحة الرباعي وما يتعلق به.
الباب الثالث: في مساحة المضلع وما يتعلق به.
الباب الرابع: في مساحة الدائرة وما يتعلق بها.
الباب الخامس: في مساحة الأجسام المستوية.
الباب السادس: في مساحة الأجسام المحدبة.
الباب السابع: في أحجام الأجسام.
الباب الثامن: في نسبة حجم الجسم إلى وزنه وعكس ذلك.
الباب التاسع: في مساحة البنيان والأبنية.

4.2 Chapter 1: On the Area of a Triangle

فصل في مساحة المثلث.

4.2.1 Section 1: Classification of Triangles

Al-Kāshī classifies triangles by their sides (equilateral, isosceles, scalene) and by their angles (acute, right, obtuse), giving geometric definitions for each.

4.2.2 Section 2: Area Formulas

Al-Kāshī presents several methods for computing the area of a triangle:

Area of a Triangle:
Method 1 — Base and Height:

$$A = \frac{1}{2} \times \text{base} \times \text{height}$$

Method 2 — Heron's Formula: For a triangle with sides a, b, c and semi-perimeter $s = \frac{a+b+c}{2}$:

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

Method 3 — Two Sides and Included Angle:

$$A = \frac{1}{2}ab\sin C$$

Example: Find the area of a triangle with sides 13, 14, 15.

$$\begin{aligned} s &= \frac{13+14+15}{2} = 21 \\ A &= \sqrt{21(21-13)(21-14)(21-15)} \\ &= \sqrt{21 \times 8 \times 7 \times 6} = \sqrt{7056} = 84 \end{aligned}$$

4.3 Chapter 4: On the Area of the Circle

فصل في مساحة الدائرة ومعرفة محيطها من قطرها وعكس ذلك.

This chapter contains one of the most celebrated passages in the *Miftāḥ*: al-Kāshī's statement about the value of π. He writes:

> Know that the circumference is three times the diameter and a fraction which is less than one seventh of the diameter, but people take it as a seventh for simplicity of calculation. Archimedes said that this fraction is smaller than a seventh and greater than ten out of seventy-one. According to what we obtained and mentioned in our article titled "al-Muḥīṭiyya" on the ratio of the diameter to the circumference, it is $3\ 8'\,29''\,44'''$ thirds after dropping fourths and what comes after, if the diameter is one. This is far more accurate than the calculation of Archimedes as we explained in the article mentioned, and it is closer to the correct value. However, nobody knows it exactly except God, the Almighty.

In modern notation:

$$\pi \approx 3 + \frac{8}{60} + \frac{29}{60^2} + \frac{44}{60^3} = 3.14159259259\ldots$$

*This value of π is accurate to about 7 decimal places. But al-Kāshī's tour de force in the Risāla al-Muḥīṭiyya extends this to **16 decimal places**, computed using a polygon with $3 \times 2^{28} = 805{,}306{,}368$ sides.*

4.3.1 Circle Area and Related Formulas

Al-Kāshī gives the standard formulas:

Circle Formulas:

$$C = \pi d = 2\pi r \qquad \text{(Circumference)}$$

$$A = \pi r^2 = \frac{\pi d^2}{4} \qquad \text{(Area)}$$

Finding diameter from circumference:

$$d = \frac{C}{\pi}$$

The Archimedean bounds: Al-Kāshī notes the classical bounds:

$$3\frac{10}{71} < \pi < 3\frac{1}{7}$$

that is:

$$\frac{223}{71} < \pi < \frac{22}{7}$$

4.4 Chapter 9: On the Measurement of Structures and Buildings

الباب التاسع في مساحة البنيان والأبنية.

This chapter is of particular interest to historians of Islamic architecture. Al-Kāshī writes:

> The practitioners of this art have only mentioned vaults and arches, and even that was not done properly. So I cover them here in a proper way among others, because the need in buildings' measurements is more in demand.

4.4.1 Section 1: On the Areas of Vaults and Arches

Al-Kāshī defines a *proper vault* (القبو الحقيقي) as a structure built on two bases between two parallel lines, consisting of five pieces: two parts of a spherical shell (or annulus or timbrel) with hollow diameter not smaller than the gap of the vault, and two other parts with larger hollow diameter.

4.4.2 Section 2: On the Area of a Dome

For a dome of diameter d and height h, al-Kāshī gives:

$$A_{\text{dome}} = 2\pi r h$$

where r is the radius of the base sphere.

4.4.3 Section 3: On the Surface Area of the Muqarnas

فصل في مساحة سطح المقرنس.

The *muqarnas* (مقرنس) is a decorative architectural element used in Islamic architecture, consisting of tiered, niched structures that create a honeycomb or stalactite effect on ceilings, corners, and the interior surfaces of domes. Al-Kāshī describes four types of muqarnas and gives formulas for calculating their surface area, so that architects could determine the quantity of materials needed.

Surface Area of a Muqarnas:
For a simple muqarnas composed of n tiers, each with average area a_i:

$$A_{\text{total}} = \sum_{i=1}^{n} a_i \times f_i$$

where f_i is a shape factor depending on the type of muqarnas cell.

This section has been studied extensively by historians of Islamic architecture. The muqarnas represents one of the most sophisticated applications of geometry in Islamic art and architecture.

5 Al-Kāshī's Calculation of π — The *Risāla al-Muḥīṭiyya*

رسالة المحيطية في إيجاد نسبة المحيط إلى القطر.

One of the most remarkable achievements in the history of mathematics is al-Kāshī's computation of π to 16 decimal places, completed in 1424 CE. This surpassed all previous records and was not improved upon in Europe for nearly two centuries, until Adriaan van Roomen (1561–1615) and later Ludolph van Ceulen (1540–1610) pushed the calculation further.

5.1 Historical Context

Al-Kāshī's motivation was astronomical: he sought to compute the circumference of the universe with such precision that the round-off error would be smaller than the width of a horse's hair. He set the accuracy of his approximation in advance and used a polygon with:

$$3 \times 2^{28} = 805{,}306{,}368 \text{ sides}$$

5.2 The Method

Al-Kāshī used the classical method of Archimedes: inscribing and circumscribing regular polygons around a circle and computing their perimeters. Starting with a hexagon and doubling the number of sides 28 times, he obtained his approximation.

Al-Kāshī's Method for Computing π:
Starting with a regular hexagon inscribed in a unit circle:

1. Initial chord: $c_0 = 1$ (side of hexagon = radius)
2. At each step, given chord c_n of the n-gon, compute chord c_{n+1} of the $2n$-gon by:

$$c_{n+1}^2 = 2 - \sqrt{4 - c_n^2}$$

   or equivalently using the recursive formula:

$$c_{n+1} = \frac{c_n}{\sqrt{2 + \sqrt{4 - c_n^2}}}$$

3. The perimeter of the inscribed N-gon is $P_N = N \cdot c_N$.
4. The approximation to π is $\pi \approx \frac{P_N}{2}$.
5. Repeat 28 times to obtain the polygon with 3×2^{28} sides.

5.3 The Result

Al-Kāshī first expressed 2π in sexagesimal notation:

$$2\pi = 6; 16, 59, 28, 1, 34, 51, 46, 14, 50$$

which is:

$$6 + \frac{16}{60} + \frac{59}{60^2} + \frac{28}{60^3} + \frac{1}{60^4} + \frac{34}{60^5} + \frac{51}{60^6} + \frac{46}{60^7} + \frac{14}{60^8} + \frac{50}{60^9}$$

He then converted this to decimal:

$$2\pi = 6.2831853071795865\ldots$$

Hence:

$$\boxed{\pi = 3.1415926535897932\ldots}$$

Source	**Date**	**Value of π**
Archimedes	~250 BCE	3.14185 (average of bounds)
Ptolemy	~150 CE	$3.14166\ldots$ $(3; 8, 30)$
Al-Kāshī	**1424 CE**	**3.1415926535897932** $\ldots$ (16 places)
Van Ceulen	1610 CE	20 places
Machin	1706 CE	100 places

Al-Kāshī's accuracy: to compute 2π with 9 sexagesimal places means the maximal error is less than $1/60^9 \approx 9.92 \times 10^{-17} < 10^{-16}$. That is, al-Kāshī computed 2π exactly up to and including the 16th decimal place.

6 Treatise V: On Algebra

المقالة الخامسة في الجبر والمقابلة.

The final treatise of the *Miftāh* is devoted to algebra (الجبر والمقابلة). Al-Kāshī presents a systematic treatment of the subject, building on the work of his predecessors — particularly al-Khwārizmī, Abū Kāmil, al-Karajī, and al-Samaw'al — while adding his own innovations.

6.1 The Chapters of Treatise V

Ch.1: On the types of equations — The six canonical forms.

Ch.2: On finding the unknown using the Rule of Double False Position (طريقة الخطأين).

Ch.3: On algebraic operations — Addition, subtraction, multiplication, and division of algebraic expressions.

Ch.4: On the extraction of roots of algebraic expressions.

Ch.5: On solving equations of higher degree — Including cubic equations.

Ch.6: On the solution of problems — Applied problems solved by algebra.

6.2 Chapter 1: The Six Canonical Forms

Al-Kāshī, following the tradition established by al-Khwārizmī, classifies all equations into six types. Using modern notation with x as the unknown and a, b as positive coefficients:

Form	Equation	Name
1	$ax = b$	Simple (مبسط)
2	$ax^2 = bx$	Squares equal roots
3	$ax^2 = b$	Squares equal numbers
4	$ax^2 + bx = c$	Squares and roots equal numbers
5	$ax^2 + c = bx$	Squares and numbers equal roots
6	$bx + c = ax^2$	Roots and numbers equal squares

Solution of $ax^2 + bx = c$ (Form 4):
Divide by a: $x^2 + \frac{b}{a}x = \frac{c}{a}$
Complete the square:

$$x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = \frac{c}{a} + \left(\frac{b}{2a}\right)^2$$

Hence:

$$\left(x + \frac{b}{2a}\right)^2 = \frac{4ac + b^2}{4a^2}$$

Solution:

$$x = \frac{-b + \sqrt{b^2 + 4ac}}{2a}$$

6.3 Chapter 2: The Rule of Double False Position

فصل في استخراج المجهول بطريقة الخطأين.

The **Rule of Double False Position** (طريقة الخطأين) is an ancient method for solving linear problems without algebra. Al-Kāshī gives a clear exposition:

Rule of Double False Position:
Given a problem of the form $ax + b = c$, make two guesses g_1 and g_2, giving results f_1 and f_2 with errors $e_1 = c - f_1$ and $e_2 = c - f_2$. Then:

$$x = \frac{g_1 e_2 - g_2 e_1}{e_2 - e_1}$$

Example: A quantity is such that when you add one-third of it to one-quarter of it, the result is 14. What is the quantity?
Guess $g_1 = 12$: $\frac{12}{3} + \frac{12}{4} = 4 + 3 = 7$, error $e_1 = 14 - 7 = 7$
Guess $g_2 = 24$: $\frac{24}{3} + \frac{24}{4} = 8 + 6 = 14$, error $e_2 = 14 - 14 = 0$

$$x = \frac{12 \cdot 0 - 24 \cdot 7}{0 - 7} = \frac{-168}{-7} = 24$$

6.4 Chapter 5: On Solving Cubic Equations

Al-Kāshī presents methods for solving cubic equations. For the equation $x^3 + px = q$, he uses an iterative method:

Iterative Solution of $x^3 + px = q$:
Rearrange as: $x = \sqrt[3]{q - px}$
Iteration:

1. Start with initial guess x_0.
2. Compute $x_{n+1} = \sqrt[3]{q - px_n}$.
3. Repeat until $|x_{n+1} - x_n| < \varepsilon$.

Example: Solve $x^3 + 2x = 20$
Start with $x_0 = 2$:

$$\begin{aligned}
x_1 &= \sqrt[3]{20 - 2 \cdot 2} = \sqrt[3]{16} \approx 2.52 \\
x_2 &= \sqrt[3]{20 - 2 \cdot 2.52} = \sqrt[3]{14.96} \approx 2.466 \\
x_3 &= \sqrt[3]{20 - 2 \cdot 2.466} = \sqrt[3]{15.068} \approx 2.472 \\
x_4 &= \sqrt[3]{20 - 2 \cdot 2.472} = \sqrt[3]{15.056} \approx 2.471
\end{aligned}$$

Solution: $\boxed{x \approx 2.471}$

This iterative method for solving cubic equations is essentially what is now known as ***fixed-point iteration****. Al-Kāshī used a similar approach in his Risāla al-Watar wal-Jayb to compute* $\sin(1^\circ)$ *to unprecedented accuracy.*

7 Tables in the *Miftāh al-Ḥisāb*

Al-Kāshī notes in his preface:

> I placed for most operations a model in tabular form so that they may be easily grasped by the intelligent. And all the tables placed in this book — "my excuse is my frankness, and its sweet and bitter are part of me" — except seven tables.

7.1 The Seven Auxiliary Tables

Table 1: Table of products of numbers below ten — A basic multiplication table.

Table 2: The Network (الشبكة) for multiplication — A lattice multiplication aid.

Table 3: Table of powers of places — Place value reference.

Table 4: Example of the agreement of roots — Root extraction reference.

Table 5: Table for knowing the rank of the product and quotient of division — Place value computation aid.

Table 6: Table of sines — Essential for astronomical calculations.

Table 7: Table for knowing the parity of the product and division — Even/odd determination.

7.2 Sample: Multiplication Table (Table 1)

×	1	2	3	4	5	6	7	8	9	10
1	1	2	3	4	5	6	7	8	9	10
2	2	4	6	8	10	12	14	16	18	20
3	3	6	9	12	15	18	21	24	27	30
4	4	8	12	16	20	24	28	32	36	40
5	5	10	15	20	25	30	35	40	45	50
6	6	12	18	24	30	36	42	48	54	60
7	7	14	21	28	35	42	49	56	63	70
8	8	16	24	32	40	48	56	64	72	80
9	9	18	27	36	45	54	63	72	81	90
10	10	20	30	40	50	60	70	80	90	100

A Biographical Note on Al-Kāshī

Jamshīd Ghiyāth al-Dīn ibn Masūd ibn Mahmūd al-Ṭabīb al-Kāshī was born in Kāshān, in central Iran, around 1380 CE. His given name was Jamshīd; his father's name was Masūd and his grandfather's name was Mahmūd. He is known as al-Kāshī (or more precisely al-Kāshānī, from Kāshān), and by his honorific title Ghiyāth al-Dīn ("the rescuer of the religion"). The epithet al-Ṭabīb ("the physician") reflects his early practice of medicine before devoting himself fully to mathematics and astronomy.

A.1 Key Dates in Al-Kāshī's Life

Date	Event
∼1380	Born in Kāshān, Iran
1407	Completed *Sullam al-Samā'* (Ladder of the Heavens)
1410–11	Wrote *Mukhtaṣar dar 'ilm-i hay'at* (Compendium of Astronomy)
1413–14	Completed *Zīj al-Khāqānī*, dedicated to Ulugh Beg
∼1420	Moved to Samarkand at Ulugh Beg's invitation
1424	Completed *Risāla al-Muḥīṭiyya* (calculation of π)
2 March 1427	**Completed *Miftāh al-Ḥisāb***
1429	Wrote *Risāla al-Watar wal-Jayb* (sin $1°$)
22 June 1429	Died in Samarkand (Ramadan 19, 832 AH)

A.2 Al-Kāshī's Other Works

1. **Sullam al-Samā'** (Ladder of the Heavens, 1407) — On resolving difficulties in determining distances and sizes of celestial bodies.
2. **Mukhtaṣar dar 'ilm-i hay'at** (Compendium of Astronomy, 1410–11) — A Persian work on astronomy.
3. **Zīj al-Khāqānī** (Khāqānī Astronomical Tables, 1413–14) — Dedicated to Ulugh Beg.
4. **Risāla al-Muḥīṭiyya** (Treatise on the Circumference, 1424) — The computation of π to 16 decimal places.
5. **Risāla al-Watar wal-Jayb** (Treatise on the Chord and Sine, 1429) — Computing $\sin(1°)$ to unprecedented accuracy.
6. **Nuzhat al-Ḥadā'iq** (Garden Promenade) — On the construction and use of the Ṭabaq al-Manāṭiq instrument.
7. **Zīj al-Tashīlāt** (Tables of Facilitation) — Simplified astronomical tables.

In the words of E.S. Kennedy: "Al-Kāshī was first and foremost a master computer of extraordinary ability, witness his facile use of pure sexagesimals, his wide application of iterative algorithms, and his sure touch in so laying out a computation that he controlled the maximum error and maintained a check at all stages."

B Testimonies of Modern Scholars

"*Key to Arithmetic* is the crowning achievement of Islamic arithmetic."

— J.L. Berggren

"It was only 1948 that Luckey presented the first extensive and detailed study of al-Kāshī's work, and the traditional historical picture was explicitly shaken."

— R. Rashed

"In the richness of its contents and in the application of arithmetical and algebraic methods to the solution of various problems, including several geometric ones, and in the clarity and elegance of exposition, this voluminous textbook [that is, *Miftāh*] is one of the best in the whole of medieval literature; it attests to both the author's erudition and his pedagogical ability. Because of its high quality the Miftāh was often recopied and served as a manual for hundreds of years."

— A.P. Youschkevitch & B.A. Rosenfeld

"It is in the work of Giyāth al-Dīn Jamshīd al-Kāshī in the early fifteenth century that we first see both a total command of the idea of decimal fractions and a convenient notation for them."

— V.J. Katz

"The power and elegance of computational algorithms developed by Islamic mathematicians of the ninth through the fifteenth centuries has only recently come to be appreciated. The most successful of these computers was the Iranian scientist, Jamshīd Ghiyāth al-Dīn al-Kāshī."

— E.S. Kennedy

"In the determination of π, and in computational mathematics as a whole, al-Kāshī was a pioneer."

— J.P. Hogendijk

"*Key to Arithmetic* was used for centuries by astronomers, architects, artisans, surveyors, and merchants as a textbook in the Islamic world."

— J. Freely

C Note on the Manuscripts and Editions

The *Miftāh al-Ḥisāb* was preserved in numerous manuscript copies, reflecting its importance and wide use. Key manuscripts include:

Manuscript	**Date**	**Location**
Nuruosmaniye 2967	854 H / 1450 CE	Istanbul, Nuruosmaniye Library
Esad Efendi 3768	882 H / 1477 CE	Istanbul, Sūleymāniye Library
Hūsrev Paṣa 256	889 H / 1484 CE	Istanbul, Sūleymāniye Library
Aya Sofya 2713	896 H / 1491 CE	Istanbul, Sūleymāniye Library
Leiden Or. 178	17th century	Leiden University Library

C.1 Printed Editions

1. **1887 Arabic edition** — The present edition, published in the Ottoman Empire. This was the first printed edition of the complete Arabic text.

2. **1951 German edition** — P. Luckey, *Die Rechnenkunst bei Gamschid b. Mas'ud al-Kasi* (Wiesbaden: Franz Steiner). The first major modern study.

3. **1960 English edition** — A. Dakhel, *The Extraction of the n-th Root in the Sexagesimal Notation* (Beirut: American University of Beirut). Study of Chapter 5, Treatise 3.

4. **1969 Arabic edition** — A.S. al-Dimirdāsh and M.H. al-Ḥifnī al-Shaykh, editors (Cairo: Dār al-Kātib al-'Arabī).

5. **1977 Arabic edition** — N. Nābulsi, editor (Damascus: University of Damascus Press).

6. **1956 Russian translation** — B.A. Rosenfeld, V.S. Segal, A.P. Youschkevitch, *Klyuch Arifmetiki* (Moscow).

The end of what has been compiled from the Miftāh al-Ḥisāb of Jamshīd al-Kāshī

تم ما انتقي من مفتاح الحساب لجمشيد الكاشي
